Name ____ Date ___________ Period _____

Transformations of f(x) = |x|, Part II

TEACHER KEY | Algebra II | Unit 2 | Day 3 | TEK A2.6.C

Objective: I can describe and graph the effect of c and d on f(x) = a|x-c|+d, and combine all four transformations into one function.

Vocabulary

Horizontal shift — f(x-c) moves the graph left/right. c>0 shifts **right**, c<0 shifts **left** - opposite the sign.

Vertical shift — f(x)+d moves the graph up/down. d>0 shifts **up**, d<0 shifts **down** - same as the sign.

Vertex, general form — for f(x) = a|x-c|+d, the vertex is always at **(c, d)**.

Part 1 — Horizontal Shift: f(x - c)

Function	Shift	Vertex
f(x) = |x - 3|	**right 3**	**(3, 0)**
f(x) = |x + 2|	**left 2 (rewrite as |x - (-2)|)**	**(-2, 0)**
f(x) = |x - 7|	**right 7**	**(7, 0)**

Part 2 — Vertical Shift: f(x) + d

Function	Shift	Vertex
f(x) = |x| + 4	**up 4**	**(0, 4)**
f(x) = |x| - 5	**down 5**	**(0, -5)**
f(x) = |x| + 1.5	**up 1.5**	**(0, 1.5)**

Part 3 — Combine All Four: f(x) = a|x - c| + d

"Compressed by 1/3, shifted down 5, shifted left 2" becomes one function:

f(x) = **(1/3)|x + 2| - 5** vertex: **(-2, -5)**

You Try. Describe f(x) = -2|x - 1| + 6, and give its vertex.

vertical stretch by 2, reflected over the x-axis, shifted right 1 and up 6 vertex: **(1, 6)**

"Stretched by 4, shifted up 2, shifted right 5" becomes one function:

f(x) = **4|x - 5| + 2** vertex: **(5, 2)**

Part 4 — Real World: Dunhill Drafting's Roofline

Roofline data was modeled by trial-and-error transformation, landing on f(x) = -1.2|x - 3| + 8.

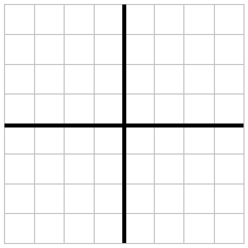
Vertex (highest point of the roof): **(3, 8)**

Describe the transformations from the parent: **stretched by 1.2, reflected over the x-axis (opens downward), shifted right 3 and up 8**

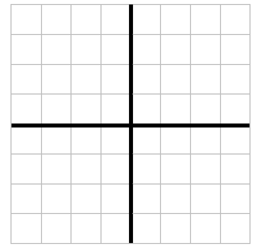
Part 5 — Graph and Compare

Graph both. Each square is 1 unit; both axes run -5 to 5.

A. $f(x) = |x - 2| - 3$



B. $f(x) = |x + 3| + 2$



Vertex of A: **(2, -3)** Vertex of B: **(-3, 2)**

Practice

1. Give the vertex of $f(x) = |x + 5| - 1$. **(-5, -1)**
2. Give the vertex of $f(x) = 4|x - 6| + 2$. **(6, 2)**
3. Write a function with vertex (-4, 7) and no stretch/reflection. **$f(x) = |x + 4| + 7$**
4. Give the vertex of $f(x) = -3|x + 8| - 6$. **(-8, -6)**
5. Write a function with vertex (2, -9) and a vertical stretch by 5. **$f(x) = 5|x - 2| - 9$**
6. Give the vertex of $f(x) = (1/4)|x - 10| + 3$. **(10, 3)**
7. A function has vertex (-6, 0) and opens downward with no stretch. Write it. **$f(x) = -|x + 6|$**
8. True or False: every function of the form $a|x - c| + d$ has the SAME shape as the parent, just moved and resized. Explain. **True - it is still two straight rays meeting at a vertex, always symmetric about a vertical line through the vertex**
9. Give the vertex of $f(x) = 2|x + 1| - 4$, and state whether it opens up or down. **(-1, -4), opens up**
10. Write a function with vertex (0, -3) that is a reflection of the parent with no stretch. **$f(x) = -|x| - 3$**

Exit Ticket

Describe every transformation of $f(x) = -(1/2)|x + 3| - 4$ from the parent, and give its vertex.

vertical compression by 1/2, reflected over the x-axis, shifted left 3 and down 4 vertex **(-3, -4)**